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Predictive Analysis

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Table of contents

1. Motivation
2. Concepts
3. Application
4. Key Message

Explanation and prediction answer different questions

Explanatory objective

- What causal relationship is of interest?
- Which assumptions identify the effect?
- How should a coefficient be interpreted?
- Which alternative explanations remain?

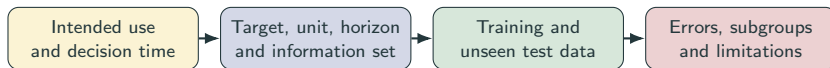
Predictive objective

- What outcome must be predicted?
- Which information is available at prediction time?
- How accurate are predictions on unseen observations?
- Where and when does performance fail?

A variable can improve prediction without causing the outcome. A causally meaningful model can still predict poorly when the future differs from its training data.

Choose the objective before choosing the model or metric.

Evaluation must reproduce the intended use



- Training performance measures how well a model fits data it has already seen.
- Test performance estimates generalization only when the test mimics future use.
- Time-dependent data usually require earlier training and later testing.
- Any information unavailable at prediction time creates leakage.

A convenient random split can answer the wrong performance question.

Define a prediction contract

Prediction target

- outcome and unit;
- target population;
- prediction horizon; and
- scale and required output.

Information set

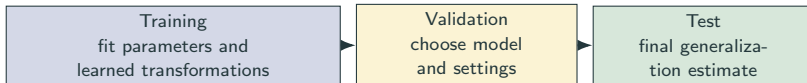
- prediction time;
- variables available then;
- data latency and revisions; and
- allowed history.

Evaluation contract

- split strategy;
- baselines and candidates;
- metrics and subgroups; and
- intended and excluded uses.

Predicting yield before planting, during the season and after harvest are different tasks because their horizons and available information differ.

Training, validation and testing have distinct roles



- Fit imputation, scaling, feature selection and model parameters on training data only.
- Use validation or resampling to compare alternatives without consuming the test set.
- Inspect the test set once the workflow is fixed; repeated tuning turns it into validation data.
- Preserve groups and temporal order when they define the deployment problem.

The test set protects an honest question only while it remains unseen.

Leakage makes performance look better than use

Common leakage routes

- future outcomes or revised future records;
- preprocessing learned from all years;
- selecting variables after test inspection;
- duplicate or related units across splits;
- predictors measured after the prediction time.

Prevent leakage through design

- timestamp every input by availability;
- split before learned preprocessing;
- fit the complete recipe on training data;
- preserve group and time boundaries;
- document every test-set interaction.

A seasonal measure can be a predictor only if it is available when the intended prediction is made. Otherwise the task is nowcasting or retrospective estimation, not an earlier forecast.

Baselines define the minimum useful performance

Candidate	Prediction rule	What it tests
Historical mean	Predict one training-period average	Can any structure beat a constant?
Country mean	Predict each country's training average	Is country identity enough?
Linear trend	Extrapolate a training-period time trend	Does common temporal change help?
Country + trend	Combine country levels and time	Does structured regression generalize?
Added predictors	Use variables available at prediction time	Does extra complexity earn its cost?

- A baseline must follow the same split and information rules as every candidate.
- More complex models must improve relevant unseen-data performance, not only training fit.
- Simplicity, stability, computation and interpretability remain deployment considerations.

Metrics summarize errors—and hide structure

Prediction error

$$e_i = y_i - \hat{y}_i$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |e_i|$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n e_i^2}$$

Interpret with context

- MAE weights absolute errors linearly;
- RMSE emphasizes larger errors;
- units follow the modeled outcome scale;
- averages can hide country and period failures;
- uncertainty grows with small test sets.

Generalization depends on future similarity. Trends, shifts, changing variance and new countries can make historical test performance optimistic for later use.

Define a prediction task before model fitting

Prediction contract

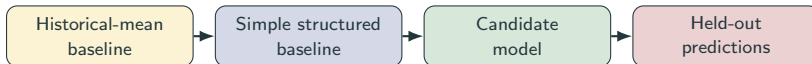
- target, unit and prediction scale;
- target population and observation grain;
- prediction time and horizon;
- development and evaluation periods;
- information available at prediction time.

Boundary

- test outcomes remain unseen during fitting;
- the evaluation represents the intended population;
- temporal prediction may require extrapolation;
- unavailable future information is excluded;
- supported and excluded uses are documented.

The split must ask whether patterns learned from available data transfer to the intended use.

Fit on development data and predict held-out observations



- Learn every transformation and parameter from development data only.
- Generate one prediction from each candidate for every held-out observation.
- Preserve observed outcomes only for evaluation after prediction.
- Save model objects, split definitions and row-level predictions reproducibly.
- Add predictors only when their availability matches the prediction time.

Key message: evaluate the complete predictive workflow

Evidence to deliver

- prediction contract and split manifest;
- row-level held-out predictions;
- MAE and RMSE overall and by relevant group;
- error plots over groups and time;
- comparison with every baseline; and
- reproducible model artifacts and report.

Limitations to state

- small test sets provide limited evidence;
- performance can differ across groups;
- future distribution shifts remain possible;
- log-scale errors need outcome-scale context;
- predictive accuracy does not imply causality.

A predictive model is useful only for a stated task when it improves on relevant baselines, survives realistic unseen-data evaluation and remains reproducible at the point of use.

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